

AWS SUMMIT



AIM201

From Demo to Deployment: Solving Agentic AI's Toughest Challenges

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Agenda

1. When to Build
2. Data
3. Trust
4. Operational Excellence
5. Reliability

Start here: Do you need to build custom?

WORKPLACE PRODUCTIVITY

Amazon Quick

Automate core workflows across sales, marketing, HR, finance, support

SOFTWARE DEVELOPMENT

Kiro

Integrate AI into coding, testing, deployment, maintenance

CUSTOMER EXPERIENCE

Amazon Connect

Deliver personalized customer experiences at scale

SECURITY & DEVOPS

Frontier Agents

Transform the way you build, secure, and operate software

Production-ready | Fully-managed | Customizable with your data | Implement in hours or days



When to build custom

Unique Business Logic | Deep System Integrations | Specialized Domain Knowledge | Full Control

AI Platform

Amazon Bedrock

Agent Development Platform

Amazon Bedrock
AgentCore &
AgentCore Harness

Agentic Framework

Strands Agents

Model & Framework-
agnostic tools

Guardrails

Observability

Governance

Customization

The prototype to production challenges

Excitement
and potential



POC

Challenges on the path to production in four pillars

Data &
Context

Trust

Operational
Excellence

Reliability

Meaningful
business value

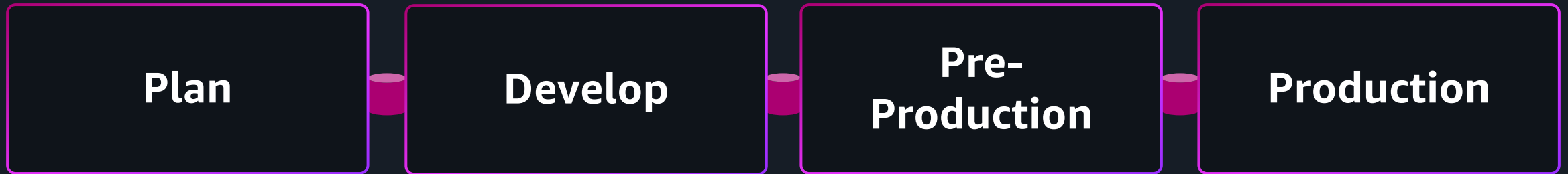


AI production
agents

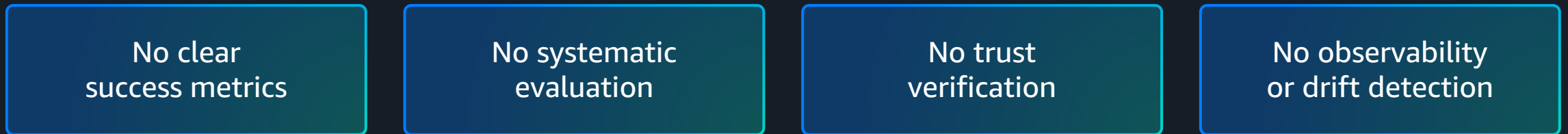


Why Agents stall: It's a lifecycle problem

The challenge isn't one gap — it's a series of stage gates



Common failure modes:



This lifecycle guides everything that follows.

PILLAR 1

Data & Context

Does your agent have the right information?

- Data Quality — Garbage in, garbage out
- Retrieval & RAG — Right context, right time
- Memory — Remember what matters
- Grounding — Prevent hallucinations

Lifecycle focus: Planning → Development

Plan



Dev



Pre-Prod



Prod



Context engineering: The #1 lever for agent quality

W — Write

Clear system prompts,
role definitions,
constraints

S — Select

Retrieve only
relevant context
(RAG, tool results)

C — Compress

Summarize,
prioritize
within token
budget

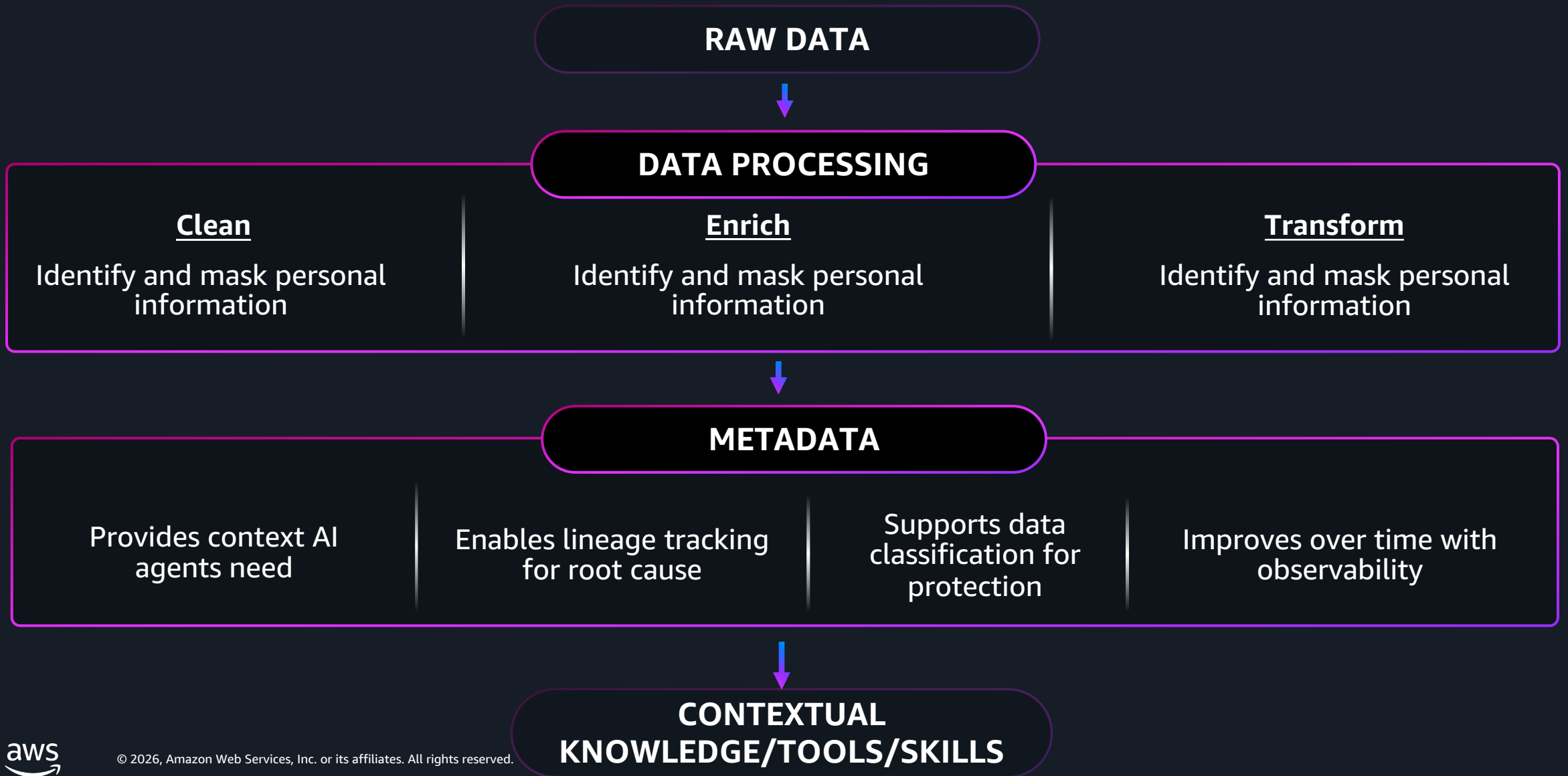
I — Isolate

Each agent gets
only its needed
context

"The model with better context outperforms a better model with bad context"

– Dr. Mel Li

Turning raw data into knowledge



Memory: Remember what matters

Short-term memory

- Conversation history
- Current session context
- Working memory

Long-term memory

- User Preferences
- Past Interactions
- Facts & Knowledge

AgentCore Memory provides both

Fully-managed

No vector DBs to run

User-namespaces

Each user's memory is isolated

Persistent

Survives across sessions

“A smart AI that forgets you after every conversation is still just a very smart stranger” – Claude Opus 4.6

PILLAR 2

Trust

Can users trust that their data is protected and the agent stays within bounds?

- Security – Protect users from each other
- Identity – Know who's asking
- Guardrails – Define boundaries
- Privacy – Handle data responsibly

Lifecycle focus: Development → Pre-Production



Security considerations for all agents

Identity

Security starts with identity.



AgentCore Identity

Policy

Enforce rules before execution at scale.



AgentCore Policy

Session Isolation

Every user gets their own environment.



AgentCore Runtime

Guardrails

Implement guardrails to maintain trust.



Bedrock Guardrail

Data Privacy: Handle Data Responsibly

Production agents handle sensitive data. Protect it.

PII Detection

Identify and mask
personal information

Data Residency

Enforce regional data
boundaries and
compliance requirements

Audit Trails

Track who accessed what,
and when

Retention Policies

Define and enforce data
lifecycle rules

AgentCore Provides

Session isolation (microVMs destroyed after use)

Identity-scoped access (users only see their data)

Full observability (audit every action)

PILLAR 3

Operational Excellence

Can you operate this in production at the scale you need?

- Observability – See what's happening
- Cost – Control what you're spending
- Scale & Latency – Perform under load

Lifecycle focus: Pre-Production → Production



Why is observability important?

You can't fix what you can't see



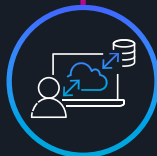
Assess system health holistically



Root cause analysis across calls



Track performance degradation or regressions



Understand and troubleshoot Agentic workflows

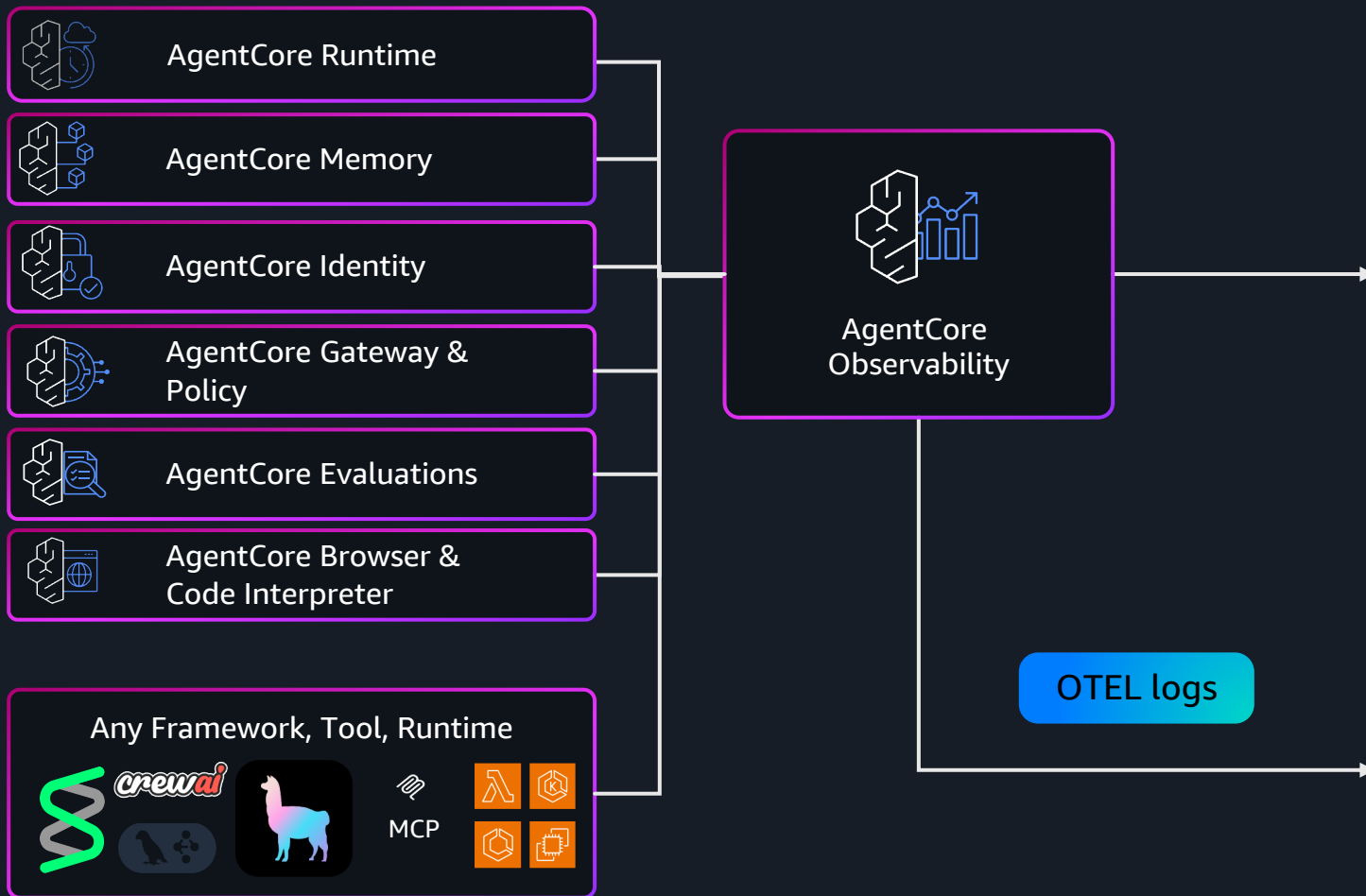
Set per-Agent cost tracking

- Per-agent, per-team cost attribution
- Real-time dashboards and budget alerts
- Cost anomaly detection (spike = possible loop)
- Chargeback and showback per business unit

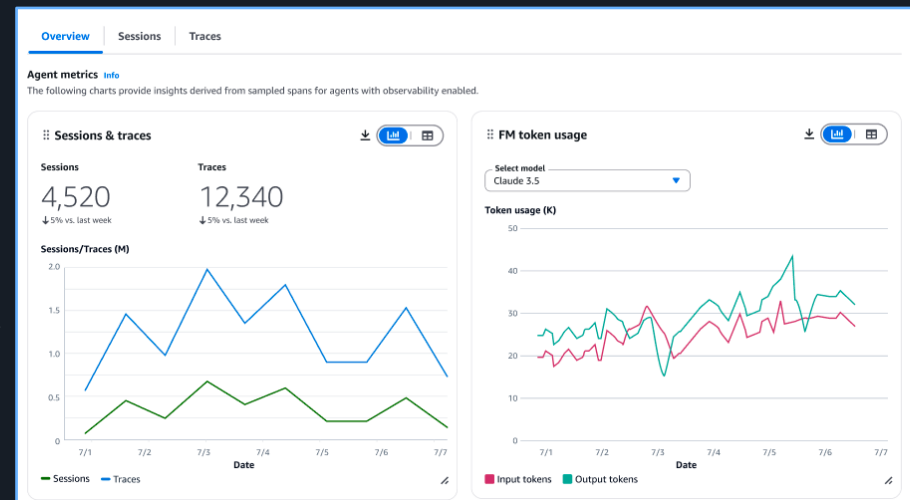
"Grupo Elfa automated R\$45M/month in procurement revenue with full agent decision traceability"



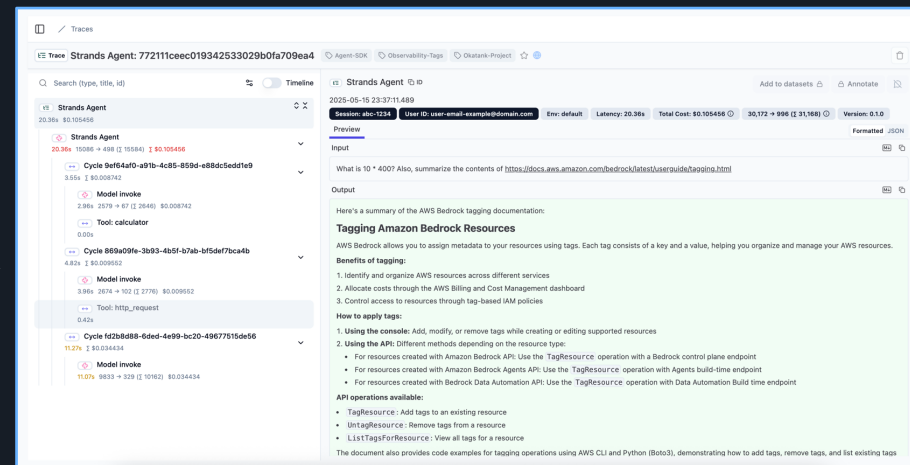
AgentCore Observability



AgentCore Observability dashboards



Third-party observability dashboards



PILLAR 4

Reliability & Evaluation

Does your agent behave consistently, and can you prove it?

- Consistency — Same input, predictable output
- Evaluations — Measure what matters
- Testing — Continuous validation
- Recovery — When things go wrong

Lifecycle focus: Every Stage – Evaluate Early, Evaluate Often



Same question, different answer?

Trust breaks when the same input
produces different outputs

Why evaluations matter



Agents are non-deterministic & user behavior evolves



Data-driven model switching



Evaluate for your specific use case, with your data

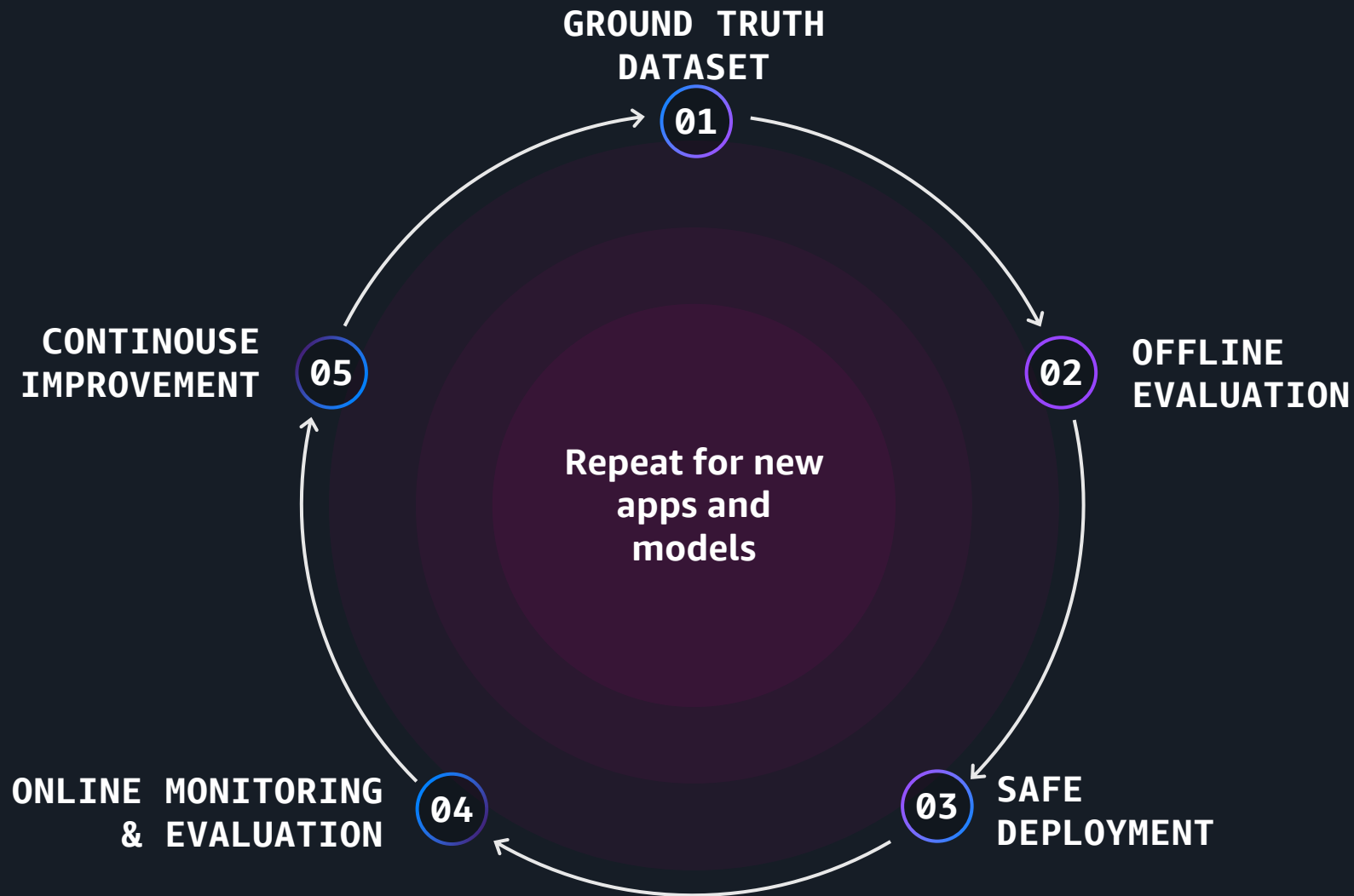


Make quality, cost, and latency tradeoffs



Monitor biases, safety, and trust

Build the evaluation feedback loop



Establish technical and business metrics and benchmarks early in development

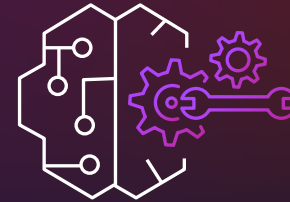
AgentCore Evaluations: Improve agent quality based on real-world performance



Real-time quality intelligence



Custom business scoring



No infrastructure overhead

Test, test and test again!

Monitor the behavior of your agent after deployment and run testing at scale

Implement continuous testing pipelines that run on every agent update

Use A/B testing to compare agent versions in production

Monitor for drift in agent behavior and model performance over time

Establish automated rollback mechanisms when quality thresholds are breached

Summary & Next Steps



4 things you can do now



DATA

Data foundation
Context engineering
Tools/Skills
Memory strategy



TRUST

Identity control
Session isolation
Guardrails
Data privacy



OPS

Observability
Cost control
Anomaly alerts



RELIABILITY

Evaluation metrics
Groundtruth dataset
Safe deployment

Start from where you are in the lifecycle. Every action compounds

Hands-On: Agent Eval & Observability Workshop

Putting it all together — from build to continuous operations





Thank you



Please complete the
session survey